

Occurrence-Resolved Physical Grouping and Prescribed-Shift Commutation on the Frontier

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Abstract

The fixed shift stored on a cut record and the fixed-shift selector after physical grouping are different mathematical objects. We formalize their relation through a commuting square. For an exact grouping matrix G with source and target shift selectors,

$$P_{\text{phys}}G = GP_{\text{src}}$$

if and only if every aggregate cross-selector block vanishes. This operator identity is still not a pathwise provenance certificate: opposite cross-shift occurrence edges can cancel in the same matrix entry. On the row-separated occurrence lift, commutation is equivalent to every nonzero occurrence edge preserving membership in the prescribed- h_0 slice. We prove this exact criterion and a sharp cancellation counterexample. Applying it to the current archive gives one positive structural result, $P_{h_0}^{\text{cut}} = I$, already supported by cut metadata intertwining. The physical grouping G , later stage matrices and downstream selector are absent, so their totality and commuting square remain NOT-TESTABLE. Treating the cut selector as a downstream selector, or aggregate cancellation as shift provenance, stops only those declared implementations. No physical cover or positive arithmetic saving is obtained.

1 Cut selection versus downstream selection

Fix a prescribed nonzero shift h_0 . TPC-136 proves that every nonzero edge through the complete cut archive retains the native shift tag [5]. If E_{cut} is its row space, define the cut coordinate selector by

$$P_{h_0}^{\text{cut}} e_c = \begin{cases} e_c, & h(c) = h_0, \\ 0, & h(c) \neq h_0. \end{cases}$$

Every row in the fixed- h_0 archive has the first case, whence

$$\boxed{P_{h_0}^{\text{cut}} = I_{E_{\text{cut}}}} \tag{1}$$

This is a literal L1 fact, not a localization estimate.

TPC-125's downstream selector is defined on later shift-tagged record spaces and must intertwine every later stage [4]. Those stages can split, aggregate or leak between shift labels. Equation (1) says nothing about an unarchived stage.

Definition 1.1 (Two selector namespaces). We write $P_{h_0}^{\text{cut}}$ only for the proved cut selector and $P_{h_0}^{\text{down}}$ only for a selector on a completed downstream archive. Equality of their names or numerical h_0 -values is not an intertwining theorem.

2 Physical occurrence grouping

Let $\mathcal{O}$ be a fully typed source occurrence set and $\mathcal{P}$ a physical occurrence set. An exact physical grouping matrix

$$G : \mathbb{C}^{\mathcal{O}} \longrightarrow \mathbb{C}^{\mathcal{P}}$$

must store, on every nonzero edge:

- (i) source occurrence, physical occurrence and physical group IDs;
- (ii) exact reconstruction multiplier;
- (iii) physical and computational multiplicities and any inverse-aggregation relation;
- (iv) masks, cover class and reconnection destination;
- (v) source and target shift tags; and
- (vi) the one inherited physical normalization.

The physical group key is not determined by a determinant label or by equality of two numerical coefficients. TPC-124 shows why a physical aggregate can hide a native residual, while TPC-117 reports that the actual growing physical map is unavailable [1, 3].

Let P_{src} and P_{phys} be the coordinate projections onto records tagged by h_0 in the source and target, respectively.

Theorem 2.1 (Aggregate block criterion). *The operator square*

$$P_{\text{phys}}G = GP_{\text{src}} \tag{2}$$

commutes if and only if

$$G_{p,o} = 0$$

whenever exactly one of $h(p), h(o)$ equals h_0 .

Proof. On a matrix entry (p, o) , the left side of (2) is

$$\mathbf{1}_{h(p)=h_0}G_{p,o},$$

while the right side is

$$G_{p,o}\mathbf{1}_{h(o)=h_0}.$$

They agree on same-selector blocks. On a cross-selector block they agree precisely when the entry vanishes. $\square$

3 Why aggregate commutation is not lineage

A grouping matrix records the sum of all transformation paths between one source and one target. Let

$$\widehat{\mathcal{E}} = \{e : o(e) \rightarrow p(e)\}$$

be the row-separated set of individual occurrence edges, with exact nonzero multipliers g_e . The corresponding path lift $\widehat{G}$ has one row for each e , so two edges never cancel inside one row.

Definition 3.1 (Occurrence-level shift preservation). The grouping is pathwise shift preserving at h_0 when every edge with $g_e \neq 0$ satisfies

$$\mathbf{1}_{h(o(e))=h_0} = \mathbf{1}_{h(p(e))=h_0}.$$

Theorem 3.2 (Row-separated commuting-square criterion). *For the row-separated occurrence lift, selector commutation is equivalent to occurrence-level shift preservation.*

Proof. The row associated with e contains one source entry g_e . The corresponding selector defect is

$$\left(\mathbf{1}_{h(p(e))=h_0} - \mathbf{1}_{h(o(e))=h_0} \right) g_e.$$

Because rows are separate, a defect in one occurrence cannot cancel a defect in another. The full defect vanishes exactly when every displayed scalar vanishes. $\square$

Proposition 3.3 (Sharp hidden-leakage example). *Aggregate commutation does not imply the pathwise condition.*

Proof. Take a source o tagged h_0 and a target p tagged $h_1 \neq h_0$. Add two occurrence edges $o \rightarrow p$, with multipliers 1 and -1 . Their aggregate matrix entry is zero, so the aggregate square commutes. Each row-separated edge crosses the selector boundary, so both fail the pathwise condition. $\square$

This is the grouping analogue of the distinction between a scalar zero defect and the row-separated matrix defect in TPC-123 and TPC-136 [2, 5]. A route that claims literal shift provenance must use the pathwise statement.

4 The current field boundary

The TPC-143 occurrence obligations require a one-to-many row-separated lift from every nonsoft cut path [6]. The required domain is

$$\mathcal{C}_X^{\text{ETO}} \sqcup \mathcal{C}_X^{\text{FUM}}.$$

No finite sample census can replace that typed domain.

For G , the current records lack physical occurrence IDs, group keys, reconstruction multipliers, multiplicity rules, cover classes and reconnection destinations. For $P_{h_0}^{\text{down}}$, they lack every later source/target shift pair and stage matrix. Consequently neither matrix in (2) can be formed.

Corollary 4.1 (Current map statuses). *The exact cut selector and the two downstream statuses are*

$$\begin{aligned} \text{H1.cut_Ph0} &= \text{PROVED}_{\text{L1}}, \\ \text{H1.frontier_G_totality} &= \text{NOT-TESTABLE}, \\ \text{H1.frontier_Ph0_downstream_totality} &= \text{NOT-TESTABLE}. \end{aligned}$$

The occurrence-level G/P_{h_0} commuting square is also NOT-TESTABLE.

5 Executable regression

The standard-library script `experiments/tpc145_group_shift_audit.py` imports the TPC-143 proof of (1). Its actual manifest contains no physical or downstream stage edge. Separate synthetic fixtures verify [theorems 2.1](#) and [3.2](#) and [proposition 3.3](#) exactly using rational multipliers.

Statement	Status	Interpretation
Cut selector $P_{h_0}^{\text{cut}}$	$\text{PROVED}_{\text{L1}}$	Identity on the current cut rows.
Aggregate block criterion	$\text{PROVED}_{\text{L0}}$	Exact finite matrix theorem.
Row-separated edge criterion	$\text{PROVED}_{\text{L0}}$	Exact path-provenance theorem.
Actual $G, P_{h_0}^{\text{down}}$	NOT-TESTABLE	Occurrence and stage edges are absent.

The audit rejects a frontier-only domain, promotion of the cut selector, fabricated physical or stage edges, and promotion of aggregate cancellation to pathwise provenance. The two stopped implementations are therefore precisely:

cut metadata as a downstream selector,
aggregate cancellation as shift provenance.

Both are scoped STOP-DECLARED-ROUTE outcomes. They do not stop a future occurrence-augmented map.

6 Claim boundary

Theorem 6.1 (Audited physical-shift conclusion). *The cut-stage identity (1) is proved, but the current source artifacts do not define a literal physical grouping or downstream fixed-shift map. Thus the selected augmented route remains open at the occurrence-lift node and is not architecture-infeasible.*

This paper proves no physical cover, quotient observability bound, frontier cancellation, localization cost, positive fixed-shift L2 estimate, endpoint below $1/400$, parity advance, prime-pair lower bound or twin-prime theorem. Determinant reserve is not charged as a physical grouping gain. The only positive application to the current archive is the exact and deliberately stage-limited identity $P_{h_0}^{\text{cut}} = I$.

References

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